

PROFESSIONAL SUMMARY

Senior RAG/Vector Database engineer with 12+ years across SaaS, property technology, nonprofit platforms, civic technology, and IT services, specializing in **Python 3.10-3.12**, **FastAPI**, **LangChain**, **pgvector**, **Pinecone**, **OpenSearch**, **AWS** for secure, scalable platforms; delivers **architecture** ownership, modernization, **performance** tuning, **test automation**, **CI/CD** discipline, and **production reliability** that improves customer workflows, **release quality**, and engineering velocity.

TECH SKILLS

- AI/ML Engineering: **Python 3.10-3.12**, **FastAPI**, **LangChain**, **pgvector**, **Pinecone**, **OpenSearch**, **AWS**, LLM application design, feature engineering, prompt/version management, model evaluation, inference APIs
- RAG & Data: Vector search, embeddings, **pgvector**, **Pinecone**, **OpenSearch**, semantic chunking, retrieval evaluation, **PostgreSQL**, S3, data validation
- Cloud & MLOps: **AWS**, **Docker**, **Kubernetes**, MLflow, **CI/CD**, model registry workflows, monitoring, cost controls, batch and real-time inference
- Application Engineering: **FastAPI**, React, **REST APIs**, **GraphQL**, auth, background jobs, **event-driven** processing, **observability**, secure integration patterns
- Senior Practices: Responsible AI guardrails, evaluation metrics, production readiness, experiment tracking, **architecture** reviews, mentoring, stakeholder alignment

PROFESSIONAL EXPERIENCE

Staff Software Engineer

Relatio

Jun 2022 - Apr 2026

Las Vegas, NV

- Led AI-enabled platform features for a SaaS workflow and analytics product using **Python 3.10-3.12**, **FastAPI**, **LangChain**, **pgvector**, **Pinecone**, **OpenSearch**, **AWS**, improving knowledge retrieval precision by **28%** through evaluated prompts, embeddings, and feedback loops.
- Designed secure inference APIs, retrieval workflows, and background processing patterns that reduced manual classification effort by **19%** across customer operations teams.
- Modernized legacy rule-based workflows with AI-assisted summarization, document extraction, and recommendation services while preserving auditability and human review controls.
- Resolved production issues involving inconsistent model outputs, slow vector searches, duplicated background tasks, token cost spikes, and incomplete trace data during peak usage.
- Implemented evaluation datasets, regression checks, prompt versioning, and CI quality gates so AI features could be released with measurable reliability expectations.
- Built **observability** around prompt inputs, retrieval quality, latency, cost, fallback rates, and user feedback to support safe incident diagnosis and product iteration.
- Partnered with **security** and product teams to define data retention, redaction, permission filtering, and guardrails for sensitive customer documents and workflow history.
- Integrated AI services with existing APIs, queues, relational data, and dashboards so product teams could add intelligent assistance without rebuilding core workflows.
- Mentored engineers on LLM application design, retrieval evaluation, production monitoring, prompt testing, and practical tradeoffs between automation and reliability.

Senior Software Engineer

AppFolio

Feb 2018 - May 2022

Goleta, CA

- Built data-driven automation features for property technology workflows using earlier versions of **Python 3.10-3.12**, **FastAPI**, **LangChain**, **pgvector**, **Pinecone**, **OpenSearch**, **AWS**, improving support triage speed by **24%** through classification and reporting enhancements.
- Implemented document, message, payment, and lease data processing services with validation, transaction logging, background jobs, and **customer-facing** status visibility.
- Migrated older heuristic logic into more testable service patterns with explicit rules, data quality checks, and regression coverage around high-volume workflows.
- Fixed defects involving duplicate notifications, delayed background jobs, inconsistent status transitions, and incorrect report outputs by adding idempotency and stronger constraints.
- Optimized SQL queries, search payloads, and cached reference data used by automation dashboards, reporting pages, and customer support workflows.
- Improved regression protection with unit, integration, and smoke tests that covered leasing, payment, notification, and reporting flows tied to automated decisions.
- Collaborated with QA and product teams to reproduce complex workflow defects, deliver targeted fixes, and avoid broad regression risk during releases.
- Mentored engineers on API design, debugging, data validation, model-adjacent evaluation thinking, and production support for **customer-facing** automation features.

Software Engineer

Blackbaud

Jul 2016 - Jan 2018

Charleston, SC

- Developed nonprofit analytics and automation workflows for fundraising, donor management, imports, reporting, and activity history using backend APIs and relational data models.
- Improved donor search and reporting relevance by tuning query logic and validation flows, reducing common large-dataset reporting delays by **21%**.

- Implemented secure data import services with duplicate detection, enrichment rules, detailed error reporting, and rollback-safe processing for support teams.
- Resolved bugs related to malformed CSV imports, timezone conversion, partial transaction failures, and incorrect donor grouping by adding stricter parsing and validation.
- Built reusable backend components for campaign records, donor profiles, activity timelines, and analytics-ready reporting flows with consistent behavior.
- Added automated tests around donation, reporting, and import workflows, improving confidence before releases and reducing regression defects.
- Worked with DevOps to improve logging, deployment checks, and operational dashboards so data-related production issues could be identified faster.

Software Developer

Granicus

May 2014 - Jun 2016

Dallas, TX

- Built civic technology automation features for agenda management, public records, document publishing, and subscription workflows with backend services and data validation.
- Improved public meeting content processing reliability through better job scheduling, retry handling, metadata extraction, and operational error reporting.
- Implemented API endpoints for agenda records, document metadata, and subscription workflows while keeping permission checks consistent across modules.
- Fixed recurring production issues involving failed document uploads, missing metadata, slow search responses, and incomplete notifications by improving validation and indexing.
- Created integration tests for workflows that handled public records, meeting documents, and notification delivery before production releases.
- Worked with support teams to investigate customer incidents, reproduce backend errors, and deliver safe patches without disrupting active government portals.

Junior Software Developer

Vology

Jun 2013 - Apr 2014

Gainesville, FL

- Supported internal automation applications using **Python** scripts, SQL, shell tools, **MySQL**, and Linux workflows for operational maintenance and reporting.
- Built small automation utilities for ticket routing, inventory updates, status classification, and operational reporting that reduced repetitive manual work.
- Fixed bugs in legacy scripts involving bad input handling, missing null checks, inconsistent database updates, and incomplete report outputs by adding validation.
- Assisted senior engineers with API maintenance, database scripts, and deployment preparation while learning production debugging and delivery practices.
- Wrote unit tests, technical notes, and troubleshooting guides for automation utilities, helping the team maintain older systems with fewer support escalations.

SELECTED PROJECTS

- Enterprise Knowledge Assistant: Built a retrieval workflow with **Python 3.10-3.12**, **FastAPI**, **LangChain**, **pgvector**, **Pinecone**, **OpenSearch**, **AWS**, embedding pipelines, document chunking, vector search, and evaluation checks that improved support answer accuracy and traceability.
- AI Workflow Automation Service: Designed inference APIs, guardrails, prompt versioning, and monitoring with **Python 3.10-3.12**, **FastAPI**, **LangChain**, **pgvector**, **Pinecone**, **OpenSearch**, **AWS**, helping product teams add AI-assisted classification and summarization without weakening **production reliability**.

EDUCATION

Arkansas State University - Bachelor's Degree in Computer Science, Sep 2009 - May 2013